

Existing Investigations Into The Compactified Two-Channel Kondo Model

The standard two-channel Kondo model,

$$H_{2\text{CK}} = J_1 \mathbf{S}_f \cdot \boldsymbol{\sigma}_1 + J_2 \mathbf{S}_f \cdot \boldsymbol{\sigma}_2 + H_{c1} + H_{c2},$$

has a stable intermediate coupling fixed point that evades a strong-coupling expansion using standard renormalised perturbation theory. This motivates a mapping where one replaces the two spin currents on distinct channels with a spin and charge current on the same channel:

$$H_{2\text{CK}} = J_1 \mathbf{S}_f \cdot \boldsymbol{\sigma} + J_2 \mathbf{S}_f \cdot \boldsymbol{\tau} + H_c$$

where $\boldsymbol{\sigma}$ and $\boldsymbol{\tau}$ are the spin and charge isospin on the first bath site. In particular, $\boldsymbol{\tau} = \sum_{\alpha,\beta} \tilde{c}_\alpha^\dagger \boldsymbol{\sigma} \tilde{c}_\beta$, where $\tilde{c}$ is the Nambu spinor:

$$\tilde{c} = \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}.$$

This gives $\boldsymbol{\tau}^\pm \equiv \frac{1}{2}(\boldsymbol{\tau}^x \pm i\boldsymbol{\tau}^y) = c_\uparrow^\dagger c_\downarrow^\dagger / c_\downarrow c_\uparrow$. Because spin and charge cannot reside at the same time, such a model cannot overscreen the impurity, and this removes the intermediate-coupling fixed point, making the strong-coupling fixed point stable.